```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

Question 1

```
# Question 1: Given a list of tuples, find the number of occurrences
of each tuple in the list.


def get_occ_tuple(lis):
  d = {}
  for item in lis:
      if not d.get(item):
          d[item] = 0
      d[item] += 1
  return d


lis = [(1, 2, 3, 4, 5), (1, 2), (1, 2), (1, 2), (1, 2), (2, 3, 4), (3,
4)]
print(get_occ_tuple(lis))
```

```
{(1, 2, 3, 4, 5): 1, (1, 2): 4, (2, 3, 4): 1, (3, 4): 1}
```

Question 2

```
# Question 2: Create a list of 10 random numbers between 1-100. Create
a balanced BST (Binary search tree) from this array

import random

class Node:
    def __init__(self, value):
        self.value = value
        self.left = None
        self.right = None

class BST:
    def __init__(self, array: list) -> None:
        self.array = array
        self.n = len(self.array)
        self.root = None

    def sort(self) -> None:
        # Bubble Sort
        n = len(self.array)

        for i in range(n):
            for j in range(n-i-1):
```

```
                if self.array[j] > self.array[j+1]:
                    self.array[j], self.array[j+1] = self.array[j+1],
self.array[j]

    def build_bst(self, array: list) -> Node:
        # Build the Balanced BST

        if not array:
            return None

        mid = (len(array))//2
        root = Node(array[mid])
        root.left = self.build_bst(array[:mid])
        root.right = self.build_bst(array[mid+1:])

        return root

    def print_preorder_bst(self, node: Node) -> None:
        # Pre-Order
        if not node:
            return

        print(node.value, end='|')
        self.print_preorder_bst(node.left)
        self.print_preorder_bst(node.right)

        return

    def direction(self) -> None:
        self.sort()
        self.root = self.build_bst(self.array)
        self.print_preorder_bst(self.root)


# Random List Generation
lis = random.sample(range(1, 100), 10)
print(f"Randomly generated list : {lis}")

obj = BST(lis)
obj.direction()

# Printing in 2D
space_count = 8
def print_in_2d(root: Node, space: int):
    if root == None:
        return
    space += space_count

    print_in_2d(root.right, space)
    print()
```

```
    print(" "*(space - space_count), end="") #
    print(root.value)
    print_in_2d(root.left, space)

print_in_2d(obj.root, 0)
```

```
Randomly generated list : [24, 69, 90, 75, 28, 96, 43, 26, 91, 10]
69|26|24|10|43|28|91|90|75|96|
                96

        91

                90

                        75

69

                43

                        28

        26

                24

                        10
```

Question 3

```
# Question 3: Given the following dictionary:Create a list of all the
keys of the dictionary, which can be formed from the values present in
the dictionary.


ldict = {
    'black': 'r',
    'hero': 'e',
    'go': 'g',
    'clue': 'i',
    'mean': 'q',
    'groan': 'o',
    'sin': 'p',
    'pint': 'u',
    'tone': 'n',
    'graze': 's',
    'sea': 't',
    'plant': 'a'
}
```

```
new_dict = {}
words_cant_be_made = []
words_can_be_made = set(ldict.keys())

for word in ldict.keys():
    for letter in word:
        if not new_dict.get(letter):
            new_dict[letter] = []
        new_dict[letter].append(word)
print("\nFirst Step")
print(new_dict)
for item in ldict.values():
    if new_dict.get(item):
        new_dict.pop(item)
print("\nSecond Step")
print(new_dict)

words_cant_be_made = []
for val in new_dict.values():
    words_cant_be_made.extend(val)
words_cant_be_made = set(words_cant_be_made)
words_can_be_made = list(words_can_be_made - words_cant_be_made)
print(f"\nWords that can be made : {words_can_be_made}")
```

```
First Step
{'b': ['black'], 'l': ['black', 'clue', 'plant'], 'a': ['black',
'mean', 'groan', 'graze', 'sea', 'plant'], 'c': ['black', 'clue'],
'k': ['black'], 'h': ['hero'], 'e': ['hero', 'clue', 'mean', 'tone',
'graze', 'sea'], 'r': ['hero', 'groan', 'graze'], 'o': ['hero', 'go',
'groan', 'tone'], 'g': ['go', 'groan', 'graze'], 'u': ['clue'], 'm':
['mean'], 'n': ['mean', 'groan', 'sin', 'pint', 'tone', 'plant'], 's':
['sin', 'sea'], 'i': ['sin', 'pint'], 'p': ['pint', 'plant'], 't':
['pint', 'tone', 'plant'], 'z': ['graze']}

Second Step
{'b': ['black'], 'l': ['black', 'clue', 'plant'], 'c': ['black',
'clue'], 'k': ['black'], 'h': ['hero'], 'm': ['mean'], 'z': ['graze']}

Words that can be made : ['sin', 'tone', 'go', 'groan', 'pint', 'sea']
```

Question 4

```
"""
Question 4: Given the following list of tuples,

a.    sort the list by 1st item in the tuple in ascending order

b.    Sort the list by 2nd item in the tuple in ascending order
```

```
c.     Repeat a, b using descending order

listA = [(1,2), (4,3), (2,10), (12, 5), (6, 7), (9,11), (15, 4)]
"""


def sort_tuple_asc(tup, choice):
    n = len(tup)
    for i in range(n):
        for j in range(n-i-1):
            if tup[j][choice] > tup[j + 1][choice]:
                tup[j], tup[j + 1] = tup[j + 1], tup[j]
    return tup

def sort_tuple_desc(tup, choice):
    n = len(tup)
    for i in range(n):
        for j in range(n-i-1):
            if tup[j][choice] < tup[j + 1][choice]:
                tup[j], tup[j + 1] = tup[j + 1], tup[j]
    return tup


listA = [(1,2), (4,3), (2,10), (12, 5), (6, 7), (9,11), (15, 4)]

def list_sort(listA):
  for i in range(2):
    choice = i
    print(f"\nAscending : {choice} element")
    print(sort_tuple_asc(listA, choice))
    print(f"\nDescending : {choice} element")
    print(sort_tuple_desc(listA, choice))

list_sort(listA)


Ascending : 0 element
[(1, 2), (2, 10), (4, 3), (6, 7), (9, 11), (12, 5), (15, 4)]

Descending : 0 element
[(15, 4), (12, 5), (9, 11), (6, 7), (4, 3), (2, 10), (1, 2)]

Ascending : 1 element
[(1, 2), (4, 3), (15, 4), (12, 5), (6, 7), (2, 10), (9, 11)]

Descending : 1 element
[(9, 11), (2, 10), (6, 7), (12, 5), (15, 4), (4, 3), (1, 2)]
```

Question 5

```
# Question 5: What do you mean by a mutable and immutable object. Are
string, list, tuple, dictionary mutable type objects or immutable
objects.
"""
Mutable : Object can be changed after it is created.
Immutable : Object can't be changed after it is created.

String : Immutable
List : Mutable
Tuple : Immutable
Dictionary : Mutable

"""
```

```
{"type":"string"}
```

Question 6

```
"""
Question 6: Learn The usage of Pandas and Numpy:

1)    Write the command to find unique values in education.
2)    Command to find the number of customers subscribed and not
subscribed.
3)    Find the mean values of all the independent variables for every
‘y’ (outcome variable)
4)    What is the mean age for every marital status
5)    Check for null values
6)    Find the descriptive statistics for each column
7)    Use pd.query(), iloc, loc and range subsetting to extract
insights from the data.
      You will be evaluated on the quality of your queries and the
rationale behind selecting them.
      You can extract upto five (5) separate insights from the data.
"""
```

```
import pandas as pd
import numpy as np

pd.options.display.max_columns = 30

data = pd.read_csv("/content/drive/MyDrive/Colab
Notebooks/prereq/banking.csv")
print(f"Shape : {data.shape}")
data.head()
```

```
Shape : (41188, 21)

   age          job  marital          education  default housing loan
\
```

```
0   44  blue-collar  married           basic.4y  unknown     yes   no

1   53   technician  married            unknown       no      no   no

2   28   management   single  university.degree       no     yes   no

3   39     services  married        high.school       no      no   no

4   55      retired  married           basic.4y       no     yes   no


    contact month day_of_week  duration  campaign  pdays  previous  \
0  cellular   aug         thu       210         1    999         0
1  cellular   nov         fri       138         1    999         0
2  cellular   jun         thu       339         3      6         2
3  cellular   apr         fri       185         2    999         0
4  cellular   aug         fri       137         1      3         1

      poutcome  emp_var_rate  cons_price_idx  cons_conf_idx  euribor3m
\
0  nonexistent           1.4          93.444          -36.1      4.963

1  nonexistent          -0.1          93.200          -42.0      4.021

2      success          -1.7          94.055          -39.8      0.729

3  nonexistent          -1.8          93.075          -47.1      1.405

4      success          -2.9          92.201          -31.4      0.869


   nr_employed  y
0       5228.1  0
1       5195.8  0
2       4991.6  1
3       5099.1  0
4       5076.2  1
```

```
# 1. Write the command to find unique values in education.
data['education'].unique()
```

```
array(['basic.4y', 'unknown', 'university.degree', 'high.school',
       'basic.9y', 'professional.course', 'basic.6y', 'illiterate'],
      dtype=object)
```

```
# 2. Command to find the number of customers subscribed and not
subscribed.
data['y'].value_counts()
```

```
0    36548
1     4640
Name: y, dtype: int64
```

```
# 3. Find the mean values of all the independent variables for every
‘y’ (outcome variable)
data.groupby(['y']).mean()
```

```
         age    duration  campaign       pdays  previous  emp_var_rate
\
y

0  39.911185  220.844807  2.633085  984.113878  0.132374      0.248875

1  40.913147  553.191164  2.051724  792.035560  0.492672     -1.233448


   cons_price_idx  cons_conf_idx  euribor3m  nr_employed
y
0       93.603757     -40.593097   3.811491  5176.166600
1       93.354386     -39.789784   2.123135  5095.115991
```

```
# 4. What is the mean age for every marital status
data.groupby(['marital'])['age'].mean()
```

```
marital
divorced    44.899393
married     42.307165
single      33.158714
unknown     40.275000
Name: age, dtype: float64
```

```
# 5. Check for null values
data.isnull().sum()
```

```
age               0
job               0
marital           0
education         0
default           0
housing           0
loan              0
contact           0
month             0
day_of_week       0
duration          0
campaign          0
pdays             0
previous          0
poutcome          0
emp_var_rate      0
```

```
cons_price_idx    0
cons_conf_idx     0
euribor3m         0
nr_employed       0
y                 0
dtype: int64
```

```
# 6. Find the descriptive statistics for each column
data.describe(include='all')
```

```
                age     job  marital          education default
housing  \
count   41188.00000   41188    41188              41188   41188
41188
unique          NaN      12        4                  8       3
3
top             NaN  admin.  married  university.degree      no
yes
freq            NaN   10422    24928              12168   32588
21576
mean       40.02406     NaN      NaN                NaN     NaN
NaN
std        10.42125     NaN      NaN                NaN     NaN
NaN
min        17.00000     NaN      NaN                NaN     NaN
NaN
25%        32.00000     NaN      NaN                NaN     NaN
NaN
50%        38.00000     NaN      NaN                NaN     NaN
NaN
75%        47.00000     NaN      NaN                NaN     NaN
NaN
max        98.00000     NaN      NaN                NaN     NaN
NaN

         loan   contact  month day_of_week      duration      campaign
\
count   41188     41188  41188       41188  41188.000000  41188.000000

unique      3         2     10           5           NaN           NaN

top        no  cellular    may         thu           NaN           NaN

freq    33950     26144  13769        8623           NaN           NaN

mean      NaN       NaN    NaN         NaN    258.285010      2.567593

std       NaN       NaN    NaN         NaN    259.279249      2.770014

min       NaN       NaN    NaN         NaN      0.000000      1.000000
```

```
25%       NaN       NaN    NaN         NaN    102.000000      1.000000

50%       NaN       NaN    NaN         NaN    180.000000      2.000000

75%       NaN       NaN    NaN         NaN    319.000000      3.000000

max       NaN       NaN    NaN         NaN   4918.000000     56.000000


               pdays      previous     poutcome  emp_var_rate
cons_price_idx  \
count   41188.000000  41188.000000        41188  41188.000000
41188.000000
unique           NaN           NaN            3           NaN
NaN
top              NaN           NaN  nonexistent           NaN
NaN
freq             NaN           NaN        35563           NaN
NaN
mean      962.475454      0.172963          NaN      0.081886
93.575664
std       186.910907      0.494901          NaN      1.570960
0.578840
min         0.000000      0.000000          NaN     -3.400000
92.201000
25%       999.000000      0.000000          NaN     -1.800000
93.075000
50%       999.000000      0.000000          NaN      1.100000
93.749000
75%       999.000000      0.000000          NaN      1.400000
93.994000
max       999.000000      7.000000          NaN      1.400000
94.767000

        cons_conf_idx     euribor3m   nr_employed             y
count    41188.000000  41188.000000  41188.000000  41188.000000
unique            NaN           NaN           NaN           NaN
top               NaN           NaN           NaN           NaN
freq              NaN           NaN           NaN           NaN
mean       -40.502600      3.621291   5167.035911      0.112654
std          4.628198      1.734447     72.251528      0.316173
min        -50.800000      0.634000   4963.600000      0.000000
25%        -42.700000      1.344000   5099.100000      0.000000
50%        -41.800000      4.857000   5191.000000      0.000000
75%        -36.400000      4.961000   5228.100000      0.000000
max        -26.900000      5.045000   5228.100000      1.000000

"""
  7. Use pd.query(), iloc, loc and range subsetting to extract
```

```
insights from the data.
  You will be evaluated on the quality of your queries and the
rationale behind selecting them.
  You can extract upto five (5) separate insights from the data.
"""
```

```
{"type":"string"}
```

```
bank = pd.read_csv('/content/drive/MyDrive/Colab
Notebooks/prereq/banking.csv')
```

```
bank.head()
```

```
   age          job  marital          education  default housing loan
\
0   44  blue-collar  married           basic.4y  unknown     yes   no

1   53   technician  married            unknown       no      no   no

2   28   management   single  university.degree       no     yes   no

3   39     services  married        high.school       no      no   no

4   55      retired  married           basic.4y       no     yes   no


    contact month day_of_week  duration  campaign  pdays  previous  \
0  cellular   aug         thu       210         1    999         0
1  cellular   nov         fri       138         1    999         0
2  cellular   jun         thu       339         3      6         2
3  cellular   apr         fri       185         2    999         0
4  cellular   aug         fri       137         1      3         1

      poutcome  emp_var_rate  cons_price_idx  cons_conf_idx  euribor3m
\
0  nonexistent           1.4          93.444          -36.1      4.963

1  nonexistent          -0.1          93.200          -42.0      4.021

2      success          -1.7          94.055          -39.8      0.729

3  nonexistent          -1.8          93.075          -47.1      1.405

4      success          -2.9          92.201          -31.4      0.869


   nr_employed  y
0       5228.1  0
```

```
1       5195.8  0
2       4991.6  1
3       5099.1  0
4       5076.2  1
```

```
## Find a person who is working as a technician and aged between 40-50
and subscribes to the bank
```

```
q1 = bank.query(
    'job == "technician"  and age >=40 and age<=50 and y == 1'
)
```

```
q1
```

```
       age         job   marital            education  default
housing  \
725     46  technician   married    university.degree  unknown
no
753     49  technician  divorced  professional.course       no
yes
809     48  technician   married             unknown       no
yes
821     45  technician   married    university.degree       no
yes
2044    49  technician   married  professional.course       no
no
...    ...         ...       ...                 ...      ...      ..
.
39244   44  technician   married    university.degree       no
no
39517   40  technician  divorced    university.degree       no
yes
39545   46  technician  divorced    university.degree  unknown
yes
40508   50  technician   married  professional.course       no
unknown
40751   48  technician   married  professional.course       no
yes

          loan    contact month day_of_week  duration  campaign  pdays
\
725        yes  telephone   jun         thu       993         6    999

753         no   cellular   jul         mon       253         3     15

809         no   cellular   jul         mon      1096         2    999

821        yes  telephone   may         tue       845         1    999

2044        no  telephone   may         fri       736         2    999
```

```
...        ...        ...   ...         ...       ...       ...    ...

39244       no   cellular   jun         wed       874         2    999

39517       no   cellular   apr         thu       335         1    999

39545       no  telephone   jul         fri      1946         4    999

40508  unknown   cellular   oct         fri       447         1      8

40751       no   cellular   aug         fri       336         8    999


       previous     poutcome  emp_var_rate  cons_price_idx
cons_conf_idx  \
725           0  nonexistent           1.4          94.465          -
41.8
753           1      success          -1.7          94.215          -
40.3
809           0  nonexistent           1.4          93.918          -
42.7
821           0  nonexistent           1.1          93.994          -
36.4
2044          0  nonexistent           1.1          93.994          -
36.4
...         ...          ...           ...             ...
...
39244         1      failure          -1.7          94.055          -
39.8
39517         0  nonexistent          -1.8          93.075          -
47.1
39545         0  nonexistent           1.4          93.918          -
42.7
40508         2      failure          -3.4          92.431          -
26.9
40751         0  nonexistent           1.4          93.444          -
36.1

       euribor3m  nr_employed  y
725        4.958       5228.1  1
753        0.870       4991.6  1
809        4.962       5228.1  1
821        4.856       5191.0  1
2044       4.857       5191.0  1
...          ...          ... ..
39244      0.715       4991.6  1
39517      1.365       5099.1  1
39545      4.962       5228.1  1
```

```
40508      0.739       5017.5  1
40751      4.964       5228.1  1

[136 rows x 21 columns]
```

```
## retrieve the data whose age between 20-30 and has a loan and
subscribes to the bank
q2 = bank.query(
    'age >= 20 and age<=30 and loan == "yes" and y == 1'
)
q2
```

```
       age            job   marital          education default housing
loan  \
404     30     technician   married  university.degree      no     yes
yes
436     29         admin.    single  university.degree      no     yes
yes
1095    29        student    single        high.school      no      no
yes
1976    29         admin.    single  university.degree      no      no
yes
2365    27    blue-collar    single           basic.9y      no     yes
yes
...    ...            ...       ...                ...     ...     ...
...
39987   29     management    single           basic.9y      no      no
yes
40384   30    blue-collar   married           basic.6y      no     yes
yes
40571   25  self-employed  divorced  university.degree      no      no
yes
40693   29         admin.    single  university.degree      no     yes
yes
41078   20        student    single           basic.4y      no     yes
yes

         contact month day_of_week  duration  campaign  pdays
previous  \
404     cellular   jul         mon       204         4    999
0
436     cellular   nov         thu       736         1    999
0
1095   telephone   may         mon       630         1    999
0
1976    cellular   jul         thu       582         2    999
0
2365    cellular   may         thu       224         4    999
0
...          ...   ...         ...       ...       ...    ...       ..
```

```
.
39987   cellular   jul         fri       968         1    999
0
40384   cellular   jul         wed      1018         2    999
0
40571   cellular   apr         mon       163         1    999
0
40693   cellular   aug         thu       254         3      3
1
41078   cellular   apr         tue       278         4      6
1

          poutcome  emp_var_rate  cons_price_idx  cons_conf_idx
euribor3m  \
404    nonexistent          -1.7          94.215          -40.3
0.870
436    nonexistent          -0.1          93.200          -42.0
4.076
1095   nonexistent           1.1          93.994          -36.4
4.857
1976   nonexistent           1.4          93.918          -42.7
4.958
2365   nonexistent          -1.8          92.893          -46.2
1.266
...            ...           ...             ...            ...
...
39987  nonexistent           1.4          93.918          -42.7
4.957
40384  nonexistent           1.4          93.918          -42.7
4.957
40571  nonexistent          -1.8          93.075          -47.1
1.405
40693      success          -2.9          92.201          -31.4
0.873
41078      success          -1.8          93.749          -34.6
0.644

       nr_employed  y
404         4991.6  1
436         5195.8  1
1095        5191.0  1
1976        5228.1  1
2365        5099.1  1
...            ... ..
39987       5228.1  1
40384       5228.1  1
40571       5099.1  1
40693       5076.2  1
41078       5008.7  1
```

```
[170 rows x 21 columns]
```

```
bank['job'].unique()
```

```
array(['blue-collar', 'technician', 'management', 'services',
'retired',
       'admin.', 'housemaid', 'unemployed', 'entrepreneur',
       'self-employed', 'unknown', 'student'], dtype=object)
```

```
##Retrieve a data whose job is technician or management and has not
subscribed to the bank
q3 = bank.query(
    'job == "technician" or job == "management" and y == 0'
)
q3
```

```
       age         job   marital            education default housing
loan  \
1       53  technician   married              unknown      no      no
no
5       30  management  divorced             basic.4y      no     yes
no
11      41  management   married    university.degree      no     yes
no
23      32  technician    single  professional.course      no      no
no
26      49  technician  divorced              unknown      no     yes
yes
...    ...         ...       ...                  ...     ...     ...
...
41169   46  technician   married             basic.9y      no      no
no
41170   32  technician    single    university.degree      no     yes
yes
41178   35  technician   married          high.school      no      no
no
41179   29  technician    single             basic.9y      no     yes
no
41186   48  technician   married  professional.course      no      no
yes

         contact month day_of_week  duration  campaign  pdays
previous  \
1       cellular   nov         fri       138         1    999
0
5       cellular   jul         tue        68         8    999
0
11      cellular   aug         thu       789         1    999
0
23      cellular   jul         thu        35         1    999
```

```
0
26      cellular   oct         thu        81         1    999
0
...          ...   ...         ...       ...       ...    ...       ..
.
41169   cellular   nov         tue       530         1    999
0
41170   cellular   aug         thu       195         3    999
0
41178  telephone   aug         fri       243         1    999
0
41179   cellular   may         mon       214         1    999
0
41186  telephone   oct         tue       200         2    999
0

          poutcome  emp_var_rate  cons_price_idx  cons_conf_idx
euribor3m  \
1      nonexistent          -0.1          93.200          -42.0
4.021
5      nonexistent           1.4          93.918          -42.7
4.961
11     nonexistent           1.4          93.444          -36.1
4.964
23     nonexistent           1.4          93.918          -42.7
4.968
26     nonexistent          -3.4          92.431          -26.9
0.754
...            ...           ...             ...            ...
...
41169  nonexistent          -0.1          93.200          -42.0
4.153
41170  nonexistent           1.4          93.444          -36.1
4.962
41178  nonexistent           1.4          93.444          -36.1
4.966
41179  nonexistent          -1.8          92.893          -46.2
1.299
41186  nonexistent          -3.4          92.431          -26.9
0.742

       nr_employed  y
1           5195.8  0
5           5228.1  0
11          5228.1  0
23          5228.1  0
26          5017.5  0
...            ... ..
41169       5195.8  0
41170       5228.1  0
```

```
41178       5228.1  1
41179       5099.1  0
41186       5017.5  0

[9339 rows x 21 columns]
```

```
##retrieve data whose marital status is not single and whose age is
less than 40 and studying in university degree
q4 = bank.loc[(bank['marital']!='single') & (bank['age']< 40) &
(bank['education']=='university.degree')]

q4
```

```
       age            job  marital          education default housing
loan  \
8       36         admin.  married  university.degree      no      no
no
54      39     management  married  university.degree      no     yes
no
62      30     management  married  university.degree      no      no
no
63      37         admin.  married  university.degree      no     yes
no
72      33   entrepreneur  married  university.degree      no      no
no
...    ...            ...      ...                ...     ...     ...
...
41109   36         admin.  married  university.degree      no     yes
no
41111   30     technician  married  university.degree      no     yes
no
41128   32  self-employed  married  university.degree      no     yes
no
41143   32         admin.  married  university.degree      no      no
no
41181   39         admin.  married  university.degree      no      no
no

         contact month day_of_week  duration  campaign  pdays
previous  \
8       cellular   jun         mon       174         1      3
1
54     telephone   may         wed        68         1    999
0
62      cellular   mar         mon       116         1    999
0
63      cellular   nov         fri       450         1    999
0
72      cellular   nov         thu        55         1    999
0
```

```
...          ...   ...         ...       ...       ...    ...       ..
.
41109   cellular   may         fri       334         3    999
0
41111   cellular   dec         thu       244         2      6
1
41128   cellular   jul         wed       181         1    999
0
41143   cellular   nov         thu        18         6    999
0
41181   cellular   nov         tue       123         2    999
0

          poutcome  emp_var_rate  cons_price_idx  cons_conf_idx
euribor3m  \
8          success          -2.9          92.963          -40.8
1.266
54     nonexistent           1.1          93.994          -36.4
4.859
62     nonexistent          -1.8          92.843          -50.0
1.811
63     nonexistent          -0.1          93.200          -42.0
4.021
72     nonexistent          -0.1          93.200          -42.0
4.076
...            ...           ...             ...            ...
...
41109  nonexistent          -1.8          92.893          -46.2
1.313
41111      success          -3.0          92.713          -33.0
0.720
41128  nonexistent           1.4          93.918          -42.7
4.963
41143  nonexistent          -0.1          93.200          -42.0
4.076
41181  nonexistent          -0.1          93.200          -42.0
4.153

       nr_employed  y
8           5076.2  1
54          5191.0  0
62          5099.1  1
63          5195.8  0
72          5195.8  0
...            ... ..
41109       5099.1  0
41111       5023.5  1
41128       5228.1  0
41143       5195.8  0
41181       5195.8  0
```

```
[3723 rows x 21 columns]
```

```
##Retrieve the data where education is highschool  and who has a loan
and who has not subscribed to the bank and aged below 25

q5 = bank.loc[(bank['education'] == 'high.school') & (bank['loan'] ==
'yes') & (bank['y'] == 0) & (bank['age'] < 25)]
q5
```

```
       age           job  marital    education  default housing
loan  \
57      24        admin.  married  high.school       no      no  yes

1171    24        admin.   single  high.school       no      no  yes

1306    24       student   single  high.school  unknown     yes  yes

2658    22        admin.  married  high.school       no     yes  yes

4133    24      services   single  high.school       no     yes  yes

6153    22       student   single  high.school       no     yes  yes

6659    20  entrepreneur   single  high.school       no      no  yes

7021    24        admin.   single  high.school       no      no  yes

7087    24        admin.   single  high.school       no     yes  yes

8635    23        admin.  married  high.school       no      no  yes

9115    24      services  married  high.school       no      no  yes

9679    20       student   single  high.school       no     yes  yes

10371   21       student   single  high.school       no     yes  yes

10974   22      services   single  high.school       no     yes  yes

11619   22       student   single  high.school       no     yes  yes

11775   20       student   single  high.school       no      no  yes

11844   24       student   single  high.school       no     yes  yes

12940   22     housemaid   single  high.school       no     yes  yes

14001   24       student   single  high.school       no      no  yes
```

14733	24	admin.	single	high.school	no	yes	yes
16267	24	student	single	high.school	no	yes	yes
17040	24	admin.	married	high.school	no	no	yes
18799	24	services	single	high.school	no	no	yes
19366	24	admin.	single	high.school	no	yes	yes
19426	24	student	single	high.school	no	yes	yes
19877	24	services	single	high.school	no	no	yes
21304	24	student	single	high.school	no	yes	yes
21405	24	services	single	high.school	no	yes	yes
21742	24	services	single	high.school	no	no	yes
22117	22	admin.	married	high.school	no	no	yes
22797	20	student	single	high.school	no	yes	yes
23036	21	student	single	high.school	no	no	yes
23390	23	blue-collar	married	high.school	no	no	yes
23405	22	student	single	high.school	no	no	yes
24398	19	student	single	high.school	unknown	no	yes
24970	24	student	single	high.school	no	yes	yes
26072	18	student	single	high.school	no	yes	yes
27861	21	services	single	high.school	no	yes	yes
32496	24	admin.	single	high.school	unknown	yes	yes
35113	22	student	single	high.school	no	no	yes
35850	24	services	single	high.school	no	yes	yes
36085	21	admin.	single	high.school	no	no	yes
36898	21	unemployed	married	high.school	no	yes	yes

```
37816   24      services   single  high.school       no     yes  yes

39048   24        admin.   single  high.school       no     yes  yes

39190   24      services   single  high.school       no     yes  yes

40406   24       student   single  high.school       no     yes  yes

40496   23      services   single  high.school       no     yes  yes

40932   23      services   single  high.school       no      no  yes


         contact month day_of_week  duration  campaign  pdays
previous  \
57     telephone   may         mon        61         1    999
0
1171    cellular   jul         tue       210         5    999
0
1306   telephone   may         fri       103         7    999
0
2658    cellular   apr         fri       156         2    999
0
4133   telephone   may         thu       198         1    999
0
6153    cellular   jun         thu       221         1    999
1
6659   telephone   may         wed       680         1    999
0
7021    cellular   mar         thu       181         1    999
0
7087    cellular   jul         thu       109         5    999
0
8635    cellular   may         thu         7         5    999
1
9115   telephone   may         fri       241         1    999
0
9679    cellular   jun         wed        74         2    999
1
10371   cellular   aug         tue       220         1    999
0
10974   cellular   jun         mon       181         1    999
0
11619  telephone   may         thu       382         3    999
1
11775   cellular   oct         mon       169         1    999
0
11844   cellular   jun         thu       344         1    999
```

0
12940 cellular jul mon 767 12 999
0
14001 cellular apr thu 157 1 999
0
14733 cellular jul wed 100 3 999
0
16267 cellular may wed 66 1 999
1
17040 telephone jun mon 1330 6 999
0
18799 telephone may fri 134 1 999
0
19366 telephone jul thu 354 1 999
0
19426 cellular nov mon 106 1 999
0
19877 cellular apr fri 1426 2 999
1
21304 cellular aug fri 44 1 999
1
21405 telephone jun mon 183 2 999
0
21742 cellular may thu 91 1 999
1
22117 cellular apr fri 108 2 999
1
22797 cellular jul tue 136 4 999
0
23036 cellular oct mon 238 1 999
0
23390 telephone jun thu 173 3 999
0
23405 cellular jul wed 349 1 999
0
24398 cellular may tue 106 4 999
0
24970 cellular aug fri 298 1 999
0
26072 cellular mar tue 103 1 999
0
27861 cellular apr thu 820 2 999
0
32496 cellular jul thu 493 1 999
0
35113 cellular jul fri 154 2 999
0
35850 cellular may fri 20 5 999
0
36085 cellular may thu 240 1 999

```
0
36898   cellular   jul         thu        85         3    999
0
37816   cellular   may         fri        69         2    999
2
39048  telephone   jul         tue       162         1    999
0
39190   cellular   may         mon       590         1    999
1
40406   cellular   jul         tue       316         3    999
1
40496   cellular   jul         mon       129         2    999
0
40932   cellular   jul         wed       207         1    999
0

          poutcome  emp_var_rate  cons_price_idx  cons_conf_idx
euribor3m  \
57     nonexistent           1.1          93.994          -36.4
4.857
1171   nonexistent           1.4          93.918          -42.7
4.961
1306   nonexistent           1.1          93.994          -36.4
4.859
2658   nonexistent          -1.8          93.075          -47.1
1.405
4133   nonexistent           1.1          93.994          -36.4
4.855
6153       failure          -2.9          92.963          -40.8
1.260
6659   nonexistent           1.1          93.994          -36.4
4.857
7021   nonexistent          -1.8          92.843          -50.0
1.584
7087   nonexistent           1.4          93.918          -42.7
4.968
8635       failure          -1.8          92.893          -46.2
1.266
9115   nonexistent           1.1          93.994          -36.4
4.859
9679       failure          -2.9          92.963          -40.8
1.260
10371  nonexistent          -2.9          92.201          -31.4
0.884
10974  nonexistent          -2.9          92.963          -40.8
1.260
11619      failure          -1.8          92.893          -46.2
1.266
11775  nonexistent          -3.4          92.431          -26.9
0.739
```

```
11844  nonexistent          -1.7          94.055          -39.8
0.729
12940  nonexistent           1.4          93.918          -42.7
4.962
14001  nonexistent          -1.8          93.075          -47.1
1.365
14733  nonexistent           1.4          93.918          -42.7
4.963
16267      failure          -1.8          92.893          -46.2
1.281
17040  nonexistent           1.4          94.465          -41.8
4.865
18799  nonexistent           1.1          93.994          -36.4
4.855
19366  nonexistent           1.4          93.918          -42.7
4.966
19426  nonexistent          -1.1          94.767          -50.8
1.050
19877      failure          -1.8          93.075          -47.1
1.405
21304      failure          -2.9          92.201          -31.4
0.849
21405  nonexistent           1.4          94.465          -41.8
4.961
21742      failure          -1.8          92.893          -46.2
1.266
22117      failure          -1.8          93.075          -47.1
1.405
22797  nonexistent           1.4          93.918          -42.7
4.961
23036  nonexistent          -3.4          92.431          -26.9
0.739
23390  nonexistent           1.4          94.465          -41.8
4.866
23405  nonexistent           1.4          93.918          -42.7
4.962
24398  nonexistent          -1.8          92.893          -46.2
1.344
24970  nonexistent          -2.9          92.201          -31.4
0.849
26072  nonexistent          -1.8          92.843          -50.0
1.687
27861  nonexistent          -1.8          93.075          -47.1
1.435
32496  nonexistent           1.4          93.918          -42.7
4.958
35113  nonexistent           1.4          93.918          -42.7
4.957
35850  nonexistent          -1.8          92.893          -46.2
1.250
```

```
36085  nonexistent          -1.8          92.893          -46.2
1.266
36898  nonexistent           1.4          93.918          -42.7
4.962
37816      failure          -1.8          92.893          -46.2
1.250
39048  nonexistent           1.4          93.918          -42.7
4.961
39190      failure          -1.8          92.893          -46.2
1.299
40406      failure          -1.7          94.215          -40.3
0.835
40496  nonexistent           1.4          93.918          -42.7
4.960
40932  nonexistent           1.4          93.918          -42.7
4.963

       nr_employed  y
57          5191.0  0
1171        5228.1  0
1306        5191.0  0
2658        5099.1  0
4133        5191.0  0
6153        5076.2  0
6659        5191.0  0
7021        5099.1  0
7087        5228.1  0
8635        5099.1  0
9115        5191.0  0
9679        5076.2  0
10371       5076.2  0
10974       5076.2  0
11619       5099.1  0
11775       5017.5  0
11844       4991.6  0
12940       5228.1  0
14001       5099.1  0
14733       5228.1  0
16267       5099.1  0
17040       5228.1  0
18799       5191.0  0
19366       5228.1  0
19426       4963.6  0
19877       5099.1  0
21304       5076.2  0
21405       5228.1  0
21742       5099.1  0
22117       5099.1  0
22797       5228.1  0
23036       5017.5  0
```

23390	5228.1	0
23405	5228.1	0
24398	5099.1	0
24970	5076.2	0
26072	5099.1	0
27861	5099.1	0
32496	5228.1	0
35113	5228.1	0
35850	5099.1	0
36085	5099.1	0
36898	5228.1	0
37816	5099.1	0
39048	5228.1	0
39190	5099.1	0
40406	4991.6	0
40496	5228.1	0
40932	5228.1	0